Depth-Dependent Directional Balance and Discrete Holonomy in Projection Tilings

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Abstract

Aperiodic projection tilings exhibit a discrete holonomy-like effect: paths that nearly close in physical space can leave nonzero residual displacement in perpendicular (address) space. We show that this residue is not explained by loop area, local step magnitude, relaxation dynamics, or dwell time at depth. Instead, it is governed by depth-dependent directional balance: every edge produces identical perpendicular-space displacement magnitude, but the directions available at each vertex vary systematically with hull depth. Deep vertices have balanced direction sets (steps cancel over random walks); shallow vertices have truncated, inward-biased sets (steps accumulate net drift). This mechanism is confirmed as genuine quasicrystal structure — not trivial boundary geometry — by null-disc controls (Spearman 0.05 vs 0.73–0.76). We further show that both Ammann–Beenker (AB) and Penrose tilings are weave-primary for identity persistence, correcting the earlier "address-led" characterisation of AB; the substrates differ not in which channel leads but in whether a working backup exists when the primary channel fails.

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1. Introduction

Aperiodic projection tilings encode information in two geometrically distinct spaces. Every vertex occupies a position in the physical tiling — parallel space — and carries a coordinate in perpendicular space, inherited from the higher-dimensional lattice point it projects. The perpendicular coordinate is not directly observable in the tiling; it is a hidden address, determined by the projection geometry, that uniquely locates each vertex within the acceptance window of the cut-and-project construction.

Li and Boyle (2024) showed that this dual encoding underpins a quantum error-correcting code: local indistinguishability among tilings of the same equivalence class guarantees that erasure of any finite patch reveals nothing about the encoded state, and that the erased region can be reconstructed from the surrounding geometry. The error model tested to date is exclusively erasure-based — errors that delete patches and produce detectable gaps. In prior work (Niedzwiecki, 2026), we introduced a diagnostic for a second error class, silent relational corruption, in

which degree-preserving scrambling of the substrate's connectivity alters global topology without producing any locally detectable trace. That study established an asymmetry between Ammann–Beenker (AB) and Penrose substrates: AB possesses a perpendicular-space address that survives complete graph destruction (identity-persistence AUC = 0.986), while Penrose does not (AUC = 0.661). Both substrates support functional self-organisation through their local relational weave; the substrates differ in whether an independent backup channel exists.

The present study asks a question that neither erasure testing nor the silent-corruption diagnostic was designed to answer: what happens to the hidden address when a path through the tiling returns to its physical starting point?

If the perpendicular-space coordinate were a simple function of physical position, any path that closed physically would close in address space as well. But in a projection tiling, vertices that are physically close can occupy very different positions in perpendicular space, because the perpendicular coordinate encodes non-local information about each vertex's relationship to the higher-dimensional lattice. A path that returns physically might, in principle, leave a residual displacement in perpendicular space — a classical, discrete analogue of the holonomy encountered in gauge theory and differential geometry, where parallel transport around a closed loop can return a vector to its starting point in a rotated state.

We find that this effect is real, reproducible, and structured. Near-closed physical paths leave perpendicular-space residue that depends not on the area enclosed by the loop — the textbook holonomy signature — but on the starting depth of the path within the perpendicular-space acceptance window. Deeper starting positions yield less residue; shallower positions yield more; and the depth-sensitivity is stronger in Penrose than in AB.

We then trace the mechanism through a sequence of progressive eliminations. An initial finding — that walkers lingering at depth recover their address more cleanly — suggests a relaxation process. Controlled experiments rule this out: the effect is instantaneous and positional, not temporal. Every edge in the tiling produces identical perpendicular-space displacement magnitude, ruling out any local stability gradient. The mechanism is instead depth-dependent directional balance: deep vertices have symmetrically distributed perpendicular-space edge directions, so random-walk contributions cancel; shallow vertices, truncated by the acceptance-window boundary, have biased direction sets that accumulate net drift. The bias is radial and inward — a geometric consequence of boundary truncation — with no chirality. Null-disc controls confirm this as genuine quasicrystal structure rather than trivial boundary geometry.

Finally, we use the mechanistic understanding to refine the channel architecture reported in prior work. Both AB and Penrose are weave-primary for identity persistence — the earlier characterisation of AB as "address-led" is corrected. The substrates differ not in which channel leads but in whether a working backup exists: when the weave is destroyed, AB's address catches the identity signal; Penrose's cannot. The

boundary of the acceptance window — where directional balance is most biased and address drift is largest — is precisely the regime where this backup asymmetry becomes visible.

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2. Substrates and hidden coordinates

2.1 Projection construction

Both substrates studied here are cut-and-project tilings: aperiodic patterns generated by projecting a higher-dimensional periodic lattice onto an irrationally oriented subspace. The projection decomposes each lattice point's position into a parallel-space (physical) component, which determines the vertex's location in the tiling, and a perpendicular-space component, which is not visible in the tiling itself but is uniquely determined by the projection geometry. An acceptance window in perpendicular space selects which lattice points contribute vertices to the tiling: a lattice point projects to a tiling vertex if and only if its perpendicular-space component falls within the window.

The Ammann–Beenker (AB) tiling has 8-fold symmetry and is generated from a Z^4 lattice via a tetragrid (de Bruijn) construction. Its acceptance window is a regular octagon in two-dimensional perpendicular space. The Penrose tiling has 5-/10-fold symmetry and is generated from a five-dimensional lattice; its perpendicular-space acceptance window comprises pentagonal regions indexed by residue class. Both tilings are constructed here using de Bruijn's multigrid method, which produces exact vertex coordinates in both parallel and perpendicular space without numerical optimisation or relaxation.

The substrates used in this study are: AB with 22,663 vertices (N = 30 multigrid parameter) and 44,126 edges; Penrose with 28,719 vertices (N = 24) and 54,695 edges. Interior subsets (75% by perpendicular-space depth) are used for primary analysis to control for boundary truncation effects, following the protocol established in Niedzwiecki (2026).

2.2 Hull depth

Each vertex's perpendicular-space coordinate places it at a specific position within the acceptance window. We define hull depth as the normalised distance from the window boundary to the window centre, scaled to [0, 1]: a vertex at the boundary has depth 0; a vertex at the centre has depth 1. Hull depth is a continuous, scalar summary of each vertex's perpendicular-space embedding.

Depth is not a dynamical variable — it is fixed by the projection geometry and does not change under any operation on the physical graph. This invariance is what makes it a candidate diagnostic: any effect that correlates with depth is responding to the projection structure, not to the local relational topology.

2.3 Edges as uniform perpendicular-space transport

A property of the cut-and-project construction that will prove central to the analysis: every edge in the tiling corresponds to a nearest-neighbour bond in the parent lattice, and every such bond has the same length. Because the projection is linear, the perpendicular-space component of every edge also has the same magnitude. In both AB and Penrose, stepping along any edge moves the walker by exactly 1.0 in perpendicular-space displacement, regardless of which edge is traversed or where in the tiling it lies.

The directions of these unit displacements, however, are not uniform across the tiling. Each edge type corresponds to a different orientation of the parent lattice bond, and the set of edge types available at a given vertex depends on the vertex's local environment — which is itself a consequence of its perpendicular-space position. This distinction between uniform magnitude and non-uniform direction is the key to the mechanism reported in §6.

2.4 Identity channels

We retain the terminology established in prior work (Niedzwiecki, 2025, 2026) for two operationally distinct information channels. The address channel comprises features derived from perpendicular-space coordinates: quantities that are fixed by the projection and invariant under any modification of the physical graph. The weave channel comprises features derived from the graph topology: local degree, neighbour degree, clustering coefficient, and second-hop connectivity — quantities that are recomputed from the current adjacency matrix and destroyed if the graph is scrambled.

These channels are revisited in §8, where we report that both substrates are weave-primary for identity persistence and that the earlier characterisation of AB as "address-led" requires correction.

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3. Physical near-closure without address closure

If perpendicular-space coordinates were simple functions of physical position, any path that returned to its starting point in the tiling would also return to its starting address. The first question this study asks is whether that guarantee holds.

3.1 Walk protocol

Random walks of 1,024 steps were run on both substrates (10 seeds, 20,000 walkers per seed). At each step, the walker moves to a uniformly random neighbour. After the walk, the physical displacement (Euclidean distance from start to finish in parallel space) and perpendicular-space displacement (Euclidean distance from start to finish in perpendicular space) are recorded. Walkers whose final physical displacement falls below a threshold — $d \leq 1$, 2, or 3 in units of median edge length — are classified

as physical near-returns. For these walkers, the perpendicular-space displacement at walk's end is the address residue: the amount of hidden-coordinate drift that remains after the body has come home.

A shuffled control permutes the perpendicular-space coordinates randomly across vertices, breaking any geometric relationship between physical and perpendicular position while preserving the graph topology. Any structure present in the real data but absent in the shuffled control is attributable to the projection geometry.

3.2 Physical near-closure does not guarantee address closure

Across both substrates and all thresholds, walkers that return physically retain substantial address residue. The effect is not noise: it is reproducible across seeds, structured by substrate, and absent in shuffled controls at tight thresholds.

The defining pattern is depth-gating. Walkers starting from deep within the perpendicular-space hull (depth > 0.5) return with markedly less address residue than walkers starting from shallow positions (depth < 0.5):

Substrate	Threshold	Deep start	Shallow start	Gap
AB	d ≤ 1	0.593	0.966	0.373
AB	d ≤ 2	0.880	1.116	0.235
AB	d ≤ 3	0.955	1.213	0.257
Penrose	d ≤ 1	0.504	0.726	0.222
Penrose	d ≤ 2	0.886	1.178	0.292
Penrose	d ≤ 3	0.985	1.396	0.411

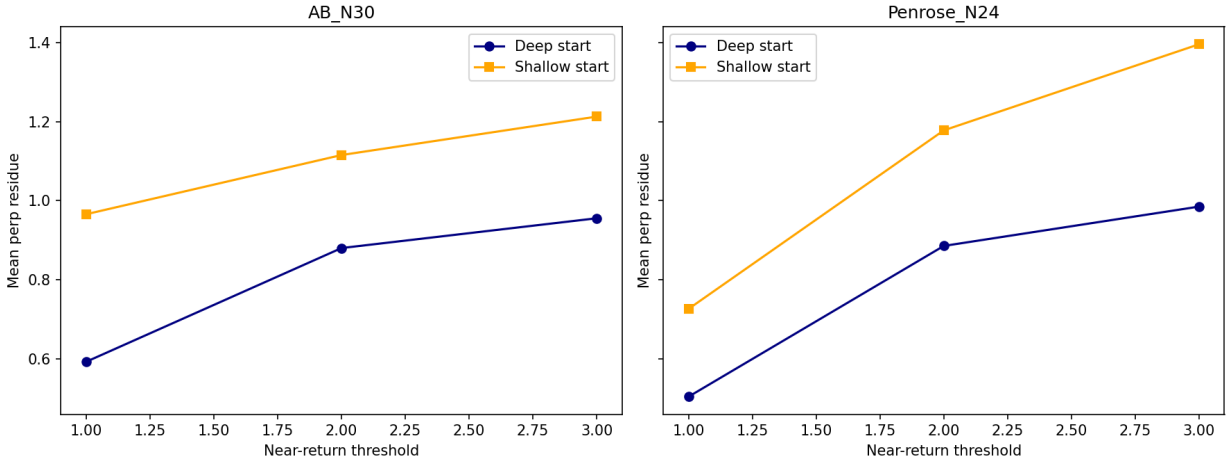


Figure 1: Address residue by start depth for physical near-returns, both substrates. Deep-start walkers consistently return with less address displacement than shallow-start walkers. The gap grows with loop size in Penrose.

Two features of this table merit attention. First, the depth gap is present in both substrates and at every threshold — this is not a boundary artifact or a single-condition result. Second, the gap grows with threshold in Penrose (0.222 at d ≤ 1 to 0.411 at d ≤ 3) but remains relatively flat in AB. Larger physical near-loops expose more of the depth-dependent structure in Penrose, consistent with its more internally differentiated perpendicular-space architecture.

3.3 The area-law form does not hold

In classical differential geometry, holonomy — the failure of parallel transport to return a vector to its original state after traversal of a closed loop — is proportional to the curvature integrated over the enclosed area. If the address residue followed this form, it would scale with the physical area enclosed by the walk path.

It does not. The Spearman correlation between loop area (computed via the shoelace formula on the walk's physical trajectory) and perpendicular-space residue is approximately 0.005 in both substrates — indistinguishable from zero. Binned means are flat. Whatever governs the address residue, it is not the area enclosed by the physical path.

This rules out the most straightforward interpretation: the address residue is not a classical Berry curvature integrated over a surface. The effect depends on where the path starts, not on what it encloses.

3.4 Shuffled controls confirm geometric origin

At the tightest threshold (d ≤ 1), walkers on the real tiling accumulate substantially more address residue than walkers on the shuffled control (AB: 0.777 vs 0.528; Penrose: 0.614 vs 0.484). The real perpendicular-space geometry produces more address drift for near-returning walkers than a random assignment of hidden coordinates to the same graph would.

At looser thresholds the real-vs-shuffled gap narrows, suggesting that the geometric structure is most consequential for the tightest near-returns — paths that come very close to home physically while failing to close in address space.

3.5 The question sharpened

The address residue is real, depth-gated, and geometry-dependent, but not area-dependent. A path that returns physically can fail to return in address space, and the severity of this failure is governed by the walker's perpendicular-space embedding depth. Deep walkers bring their address home; shallow walkers do not.

This raises the mechanistic question that occupies the remainder of the paper: why does depth predict address residue? What is it about the geometry at different hull depths that causes random-walk steps to cancel or accumulate in perpendicular space?

The next two sections pursue an initially plausible answer — that depth provides a stable zone where addresses relax over time — and show that it is wrong.

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4. Depth recovery and the linger-time interpretation

The depth-gating result in §3 establishes that perpendicular-space embedding depth predicts address residue. But start depth is a static property of the walker's origin — it says nothing about what happens along the path. If the mechanism is purely about starting position, then the walk itself is irrelevant; if something about the walk's trajectory through different depth zones also matters, the mechanism is richer than a simple initial-condition effect. This section examines path-level predictors and finds one that appears to explain the depth effect dynamically. The next section shows that this explanation is wrong.

4.1 Dip-and-return walkers recover their address

Using the same walk ensemble as §3 (10 seeds, 20,000 walkers per seed, 1,024 steps), we characterise each walker's depth trajectory: the minimum hull depth reached, the maximum excursion below start depth, and whether the walker qualifies as a "dip-and-return" — defined as reaching a depth at least one threshold unit below its start depth and finishing within one threshold unit of its start depth.

The result is counterintuitive. Walkers that dip into shallow territory and return to depth finish with less address residue than walkers that never dip:

Substrate	Threshold	Dip-return	No-dip	Gap
AB	$d \leq 1$	0.413	1.369	−0.956
AB	$d \leq 2$	0.869	1.140	−0.272
AB	$d \leq 3$	0.957	1.213	−0.256
Penrose	$d \leq 1$	0.367	1.198	−0.831
Penrose	$d \leq 2$	0.883	1.214	−0.331
Penrose	$d \leq 3$	1.010	1.388	−0.378

A naive curvature-accumulation model would predict the opposite: walkers traversing more of the tiling, passing through regions of varying depth, should accumulate more perturbation and finish with higher residue. Instead, the excursion is beneficial. Among path-level predictors, excursion depth — how far below start depth the walker dipped — is the strongest (Spearman -0.25 to -0.36), while depth range and cumulative depth change show near-zero correlation.

The pattern admits a natural reading. The deep hull is a stable zone for addresses. Non-dippers remain in the shallow region where addresses are unstable; dip-and-return walkers visit the shallow zone but come home to depth, where their address

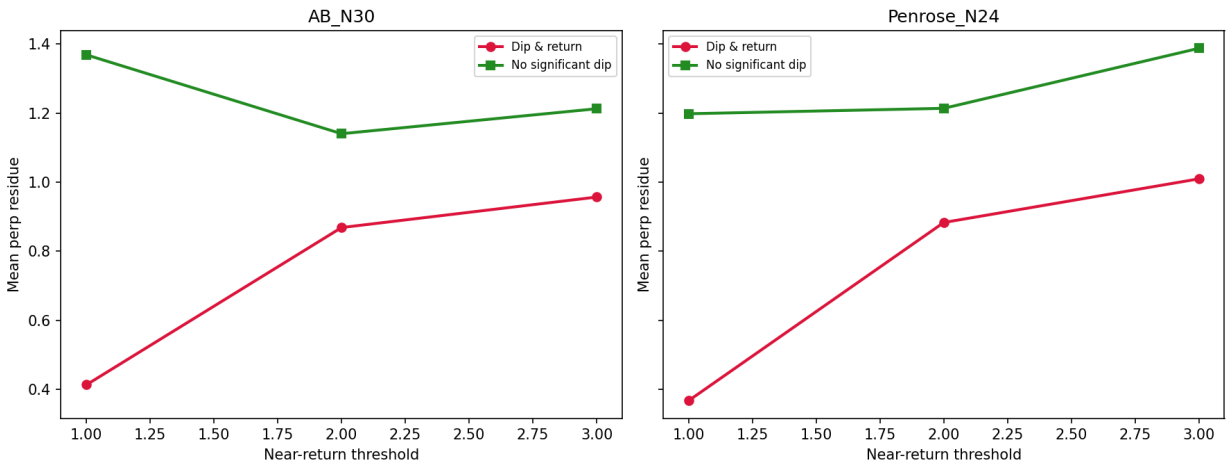


Figure 2: Address residue for dip-and-return walkers versus non-dippers, both substrates. At every threshold, walkers that venture into shallow territory and return to depth finish with substantially less address residue than walkers that remain shallow throughout.

is restored. The excursion is not damage — it is a round trip that ends in the right place.

4.2 The shape of the dip matters — but not the way expected

If the round-trip interpretation is correct, the details of the excursion trajectory might matter. We tested seven shape metrics for each walker's depth profile: asymmetry (fast descent vs slow descent), sharpness (V-dip vs U-dip), timing (early vs late dip), descent and ascent smoothness, excursion depth, and recovery fraction.

The brachistochrone hypothesis — that a fast drop and slow return, analogous to a least-time curve, would optimise address recovery — is not supported. Asymmetry has near-zero correlation with address residue (Spearman 0.02 in AB, −0.01 in Penrose). The speed of the descent is irrelevant.

What does predict residue is sharpness: V-shaped dips (brief contact with the minimum depth) produce substantially more residue than U-shaped dips (sustained time near the minimum):

Substrate	Threshold	V-dip residue	U-dip residue	Gap
AB	d ≤ 2	1.133	0.790	0.343
AB	d ≤ 3	1.231	0.894	0.337
Penrose	d ≤ 2	1.197	0.801	0.395
Penrose	d ≤ 3	1.423	0.926	0.497

Sharpness is the strongest shape-level predictor (Spearman 0.22–0.29 across condi-

tions). Penrose shows a larger U/V gap than AB (0.40–0.50 vs 0.34), consistent with its more structured centre-versus-edge distinction in perpendicular space.

4.3 The emerging interpretation

At this point in the analysis, the evidence points toward a coherent dynamical picture. The deep hull acts as a basin of attraction for addresses: walkers that spend extended time there recover their hidden coordinates, while walkers that merely brush past the minimum do not. The mechanism appears to be temporal — a relaxation process in which the address gradually settles toward a stable configuration, requiring a characteristic dwell time to take effect. The relationship between linger duration and recovery follows a smooth S-curve, with the steepest improvement occurring between roughly 10 and 80 steps at depth, and the effect is approximately twice as strong in Penrose as in AB.

This interpretation is intuitive, consistent with all the data presented so far, and wrong.

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5. Linger duration has no causal effect

The interpretation developed in §4 makes a testable prediction: if time at depth is the mechanism, then forcing walkers to spend more time at depth should produce lower address residue, all else equal. This section reports a controlled experiment that isolates linger duration from starting depth. The prediction fails completely.

5.1 Controlled walk protocol

The natural random walks in §3–4 confound linger duration with starting depth: walkers that linger longer at depth also tend to start deeper. To separate these variables, we constructed a five-phase walk protocol in which the only experimentally varied quantity is the number of forced linger steps at minimum depth:

1. **Start**: vertex sampled from the medium-depth band (hull depth 0.35–0.65), controlling initial position.
2. **Disturb**: 100 unconstrained random-walk steps, scrambling the address.
3. **Descend**: 100 steps biased toward deeper vertices, driving the walker toward the hull centre.
4. **Linger**: exactly N steps constrained to remain near the current depth (N = 0, 5, 10, 20, 40, 80, or 160).
5. **Ascend**: 200 steps biased toward shallower vertices, returning the walker toward its starting depth.

Every walker follows the same trajectory through phases 1–3 and 5. The only variable is N. If the linger-time interpretation is correct, increasing N should produce mono-

tonically decreasing final address residue. Ten seeds of 5,000 walkers each were run on both substrates.

5.2 The result is a flat line

Linger steps	AB final residue	Penrose final residue
0	1.254	1.340
5	1.258	1.335
10	1.256	1.335
20	1.253	1.335
40	1.254	1.339
80	1.255	1.338
160	1.252	1.338

Total change from 0 to 160 forced steps at depth: 0.002 in AB, 0.006 in Penrose. The linger-time interpretation is not approximately wrong or weakly supported — it is ruled out. Spending 160 steps at depth produces the same address residue as spending none.

5.3 The phase trajectory reveals the real structure

The flat line tells us what does not work. The phase-by-phase trajectory tells us what does.

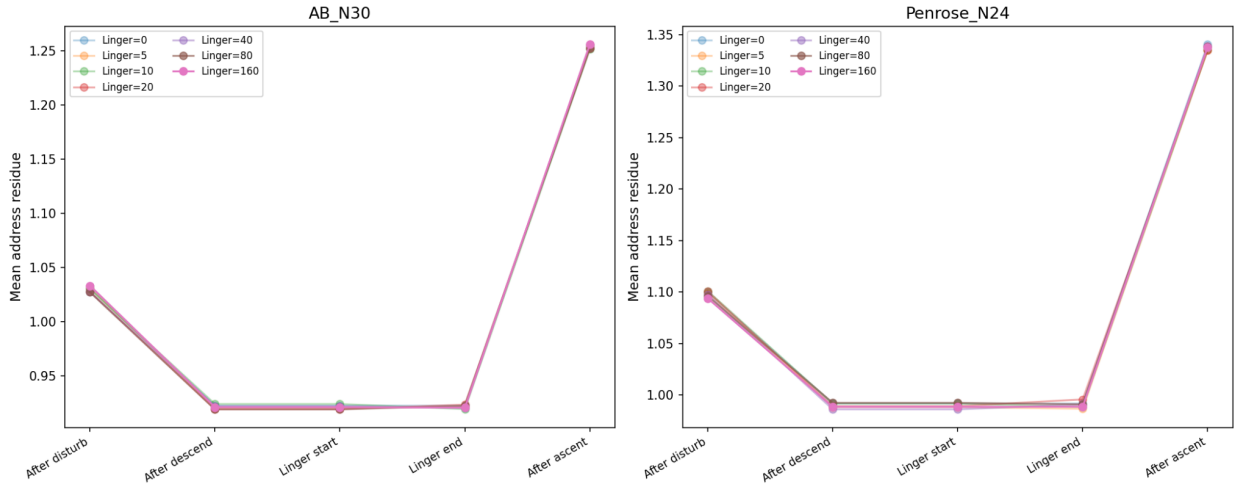


Figure 3: Mean address residue through each walk phase, both substrates. All linger durations (N = 0 to 160) are overlaid; the curves are indistinguishable. Residue drops during descent, remains flat during linger regardless of duration, and jumps during ascent.

Tracking the mean address residue at the end of each phase:

- **After disturb** (100 free steps): residue ≈ 1.03 (AB), 1.10 (Penrose). The address has been scrambled to a baseline level.
- **After descent** (100 biased-to-deep steps): residue drops to ≈ 0.92 (AB), 0.99 (Penrose). Moving toward depth immediately improves the address.
- **During linger** (N biased-to-deep steps): residue is unchanged. No recovery, no degradation, regardless of N. The address does not relax.
- **After ascent** (200 biased-to-shallow steps): residue jumps to ≈ 1.25 (AB), 1.34 (Penrose). Moving away from depth immediately worsens the address.

The address residue responds to movement between depth zones, not to time spent within one. Descending lowers residue; ascending raises it; lingering does nothing. The effect is instantaneous and positional.

5.4 What this eliminates

The controlled experiment rules out three related interpretations that the natural-walk data in §4 appeared to support:

The linger-time interpretation — that the deep hull acts as a relaxation basin where addresses gradually settle — is directly contradicted. If relaxation occurred, the linger phase would show monotonically decreasing residue with increasing N. It does not.

The S-curve relationship between linger duration and recovery observed in natural walks (§4.3) is a confound. Walkers that lingered longer at depth in uncontrolled walks also started deeper; starting depth, not linger duration, was driving the correlation.

The U-dip advantage over V-dips (§4.2) is reinterpreted. U-dip walkers do not recover better because they linger — they appear to linger because they are deeper on average, and being deeper means lower residue at every instant.

What survives is simpler than what was eliminated. Address residue is a property of the walker's current perpendicular-space position, not of its history at that position. The question is no longer "why does time at depth help?" — it is "why does position at depth help?" The answer requires examining the local geometry, which is the subject of the next section.

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6. Directional balance

The controlled experiment in §5 established that address residue is positional, not temporal: it responds to the walker's current depth, not to how long the walker has been there. The mechanism must therefore be a property of the geometry at each depth — something about the local environment that causes perpendicular-space steps to cancel at depth and accumulate near the boundary.

6.1 Every edge is locally identical

Before seeking what varies with depth, we confirmed what does not. As noted in §2.3, every edge in a cut-and-project tiling corresponds to a nearest-neighbour bond in the parent lattice, and the perpendicular-space displacement magnitude is therefore uniform across all edges. Direct measurement confirms this: the mean perpendicular-space displacement per edge is 1.000 in both substrates, with variation at the 10^{-11} level — floating-point noise, not geometric signal.

The correlation between hull depth and mean neighbour displacement is effectively zero (Spearman = 0.002 in AB, −0.075 in Penrose). There is no local stability landscape: no region of the tiling where individual steps are smaller, gentler, or more conservative in perpendicular space. A vertex at the hull centre is locally identical to a vertex at the hull boundary, measured by the magnitude of available perpendicular-space displacements.

This result eliminates local transport properties as the mechanism. If every step moves the walker by exactly 1.0 in perpendicular space regardless of location, then the depth effect reported in §§3–5 cannot arise from differences in step size. It must arise from how those uniform steps combine over the course of a walk — which is a question about direction, not magnitude.

6.2 Directional balance varies with depth

Every edge displaces the walker by 1.0 in perpendicular space, but in a specific direction determined by the edge's orientation in the parent lattice. A vertex with k neighbours has k available perpendicular-space displacement vectors, each of unit magnitude but potentially different direction. The directional balance of a vertex is measured by the magnitude of the resultant of its normalised perpendicular-space edge vectors: a resultant near zero indicates that the available directions are evenly distributed (a step in any direction is offset by the availability of a step in the opposite direction), while a large resultant indicates systematic bias.

The correlation between hull depth and directional balance is the strongest found in this study:

Depth zone	AB resultant	Penrose resultant
Shallow (< 0.2)	0.78	0.83
Mid (0.4–0.6)	0.66	0.67
Deep (> 0.8)	0.20	0.10

Spearman correlation between depth and resultant magnitude: −0.72 (AB), −0.78 (Penrose).

Deep vertices have nearly balanced perpendicular-space direction sets: a random step from a deep vertex is roughly equally likely to go in any direction in perpendicu-

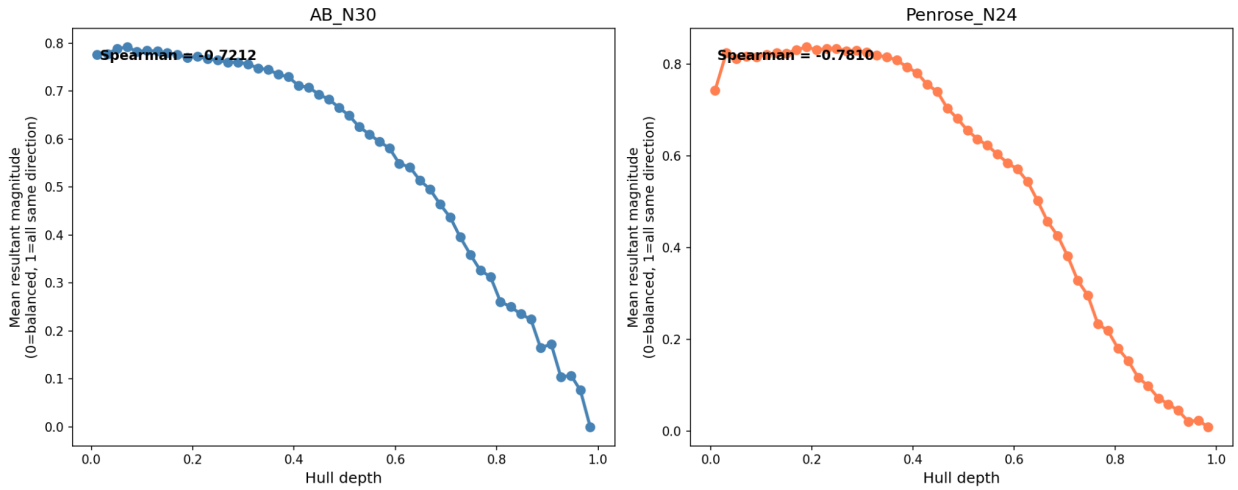


Figure 4: Directional balance (resultant magnitude) versus hull depth, both substrates. Deep vertices have near-zero resultant (balanced directions); shallow vertices have high resultant (biased directions). Spearman -0.72 (AB), -0.78 (Penrose).

lar space, so over many steps the displacements tend to cancel and the walker stays near its starting address. Shallow vertices have heavily biased direction sets: a random step from a shallow vertex has a preferred direction in perpendicular space, so displacements accumulate over a walk and the walker drifts from its starting address.

This is why depth predicts address residue. It is not about stability, relaxation, or dwell time. It is about the directional symmetry of the perpendicular-space transport options available at each hull depth.

6.3 Directional balance as a walk-residue predictor

If directional balance is the mechanism, it should predict walk-level address residue at least as well as depth does. To test this, we ran a validation ensemble (5 seeds, 10,000 walkers per seed) and computed Spearman correlations between each walker's start-vertex properties and its final address residue:

Predictor	AB d ≤ 2	AB d ≤ 3	Penrose d ≤ 2	Penrose d ≤ 3
Depth	−0.30	−0.35	**−0.42**	**−0.42**
Resultant (balance)	**0.34**	**0.35**	0.35	0.38
Net drift	0.33	0.34	0.34	0.37
Direction spread	−0.34	−0.35	−0.35	−0.38

In AB, directional balance is a stronger predictor of walk residue than depth itself (0.34 vs 0.30 at d ≤ 2). In Penrose, depth retains an advantage (0.42 vs 0.35), suggesting that hull depth captures additional information beyond directional balance alone — consistent with Penrose's more complex perpendicular-space architecture,

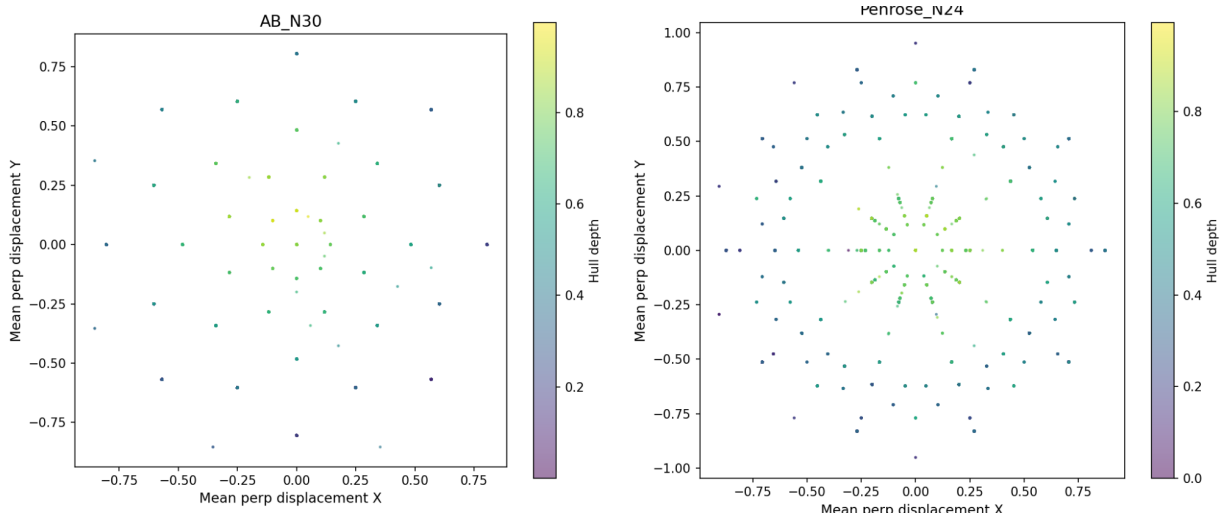


Figure 5: Mean perpendicular-space drift vector for each vertex, coloured by hull depth. Deep vertices (yellow) cluster near the origin - their neighbours' displacement directions cancel. Shallow vertices (purple) scatter far from the origin - their direction sets are systematically biased.

in which depth encodes structural relationships not fully summarised by the first-neighbour direction set.

All four predictors — depth, resultant, net drift, direction spread — fall in the same range (0.30–0.42) and are mutually correlated, consistent with their measuring overlapping aspects of the same underlying geometric structure. The directional-balance interpretation does not replace depth as a diagnostic; it explains why depth works.

6.4 The mechanism

The complete picture can now be stated. Aperiodic projection tilings have uniform local perpendicular-space transport: every step moves the walker by the same amount. But the directions available for that transport vary systematically with hull depth. Deep within the acceptance window, the projection geometry provides a balanced set of perpendicular-space directions — steps cancel over random walks, and addresses are preserved. Near the acceptance-window boundary, the available directions are truncated and biased — steps accumulate, and addresses drift.

This is a discrete, topological holonomy: uniform local transport, with non-trivial global accumulation determined by the path's position in the perpendicular-space hull. It is not Berry curvature (there is no area dependence), not a relaxation process (there is no temporal component), and not a local stability gradient (every edge is identical). It is a consequence of how the acceptance-window boundary shapes the direction sets of nearby vertices.

The next section characterises the direction and confirms the structural origin of this bias.

7. Radial drift and structural confirmation

Section 6 established that shallow vertices have biased perpendicular-space direction sets. Two questions remain. First, in what direction is the bias — does the drift point outward toward the acceptance-window boundary, inward toward the centre, or tangentially? Second, is the depth-dependent directional balance a genuine property of quasicrystalline geometry, or could any bounded set of points with a middle and an edge produce the same pattern?

7.1 The bias is inward

For each vertex, we compute the mean perpendicular-space displacement vector across all its edges — the drift vector — and decompose it into a radial component (along the line from the hull centroid to the vertex) and a tangential component (perpendicular to that line). Positive radial drift is outward; negative is inward.

The result is unambiguous. Across both substrates, the drift is overwhelmingly inward: shallow vertices (depth < 0.3) show mean radial drift of -0.753 (AB) and -0.777 (Penrose); deep vertices (depth > 0.7) show weaker but still inward drift of -0.347 and -0.251. No depth bin in either substrate has net outward drift. The correlation between depth and radial drift magnitude is strong (Spearman $= +0.73$ in AB, $+0.76$ in Penrose): shallower vertices are pulled more strongly inward.

The tangential component is zero. Mean tangential drift is 0.000132 (AB) and -0.000060 (Penrose) — indistinguishable from noise. There is no chirality, no rotational handedness, in the drift structure. The bias is purely radial.

This makes geometric sense. A vertex near the boundary of the acceptance window has most of its neighbours located further from the boundary — toward the centre — because there is simply less space outward. The available perpendicular-space edge directions are truncated on the boundary side and intact on the interior side, producing a net inward resultant. The shallower the vertex, the more severe the truncation, and the stronger the inward pull.

The drift is centripetal, not centrifugal. A walker starting at the boundary does not escape outward; it is dragged inward by the geometry of available directions. But this inward drag is precisely the mechanism of address instability: the walker is pulled away from its starting position in perpendicular space, accumulating displacement from its original address. The instability of shallow addresses is not about drifting toward the edge — it is about being displaced from wherever the walker began.

7.2 Structural confirmation: the null-disc control

The depth-dependent directional balance reported in §6 could, in principle, arise from trivial boundary geometry. Any finite set of points with a convex hull will have bound-

ary points whose Delaunay neighbours are biased toward the interior. If the quasicrystal's directional-balance structure were merely an instance of this generic boundary effect, the entire mechanism would reduce to an uninteresting property of bounded domains.

To test this, we constructed null-disc controls: sets of uniformly random points matched in number and bounding shape to each substrate, connected by Delaunay triangulation. These null discs have the same hull geometry, the same density, and the same kind of boundary — but no quasicrystalline order. If the directional-balance effect is generic, it should appear in the null discs. If it is specific to the projection geometry, it should not.

The result is definitive:

Substrate	Real tiling	Null disc	Shuffled perp
AB	0.728	0.052	0.615
Penrose	0.758	0.046	0.713

(Values are Spearman correlation between hull depth and directional-balance resultant magnitude.)

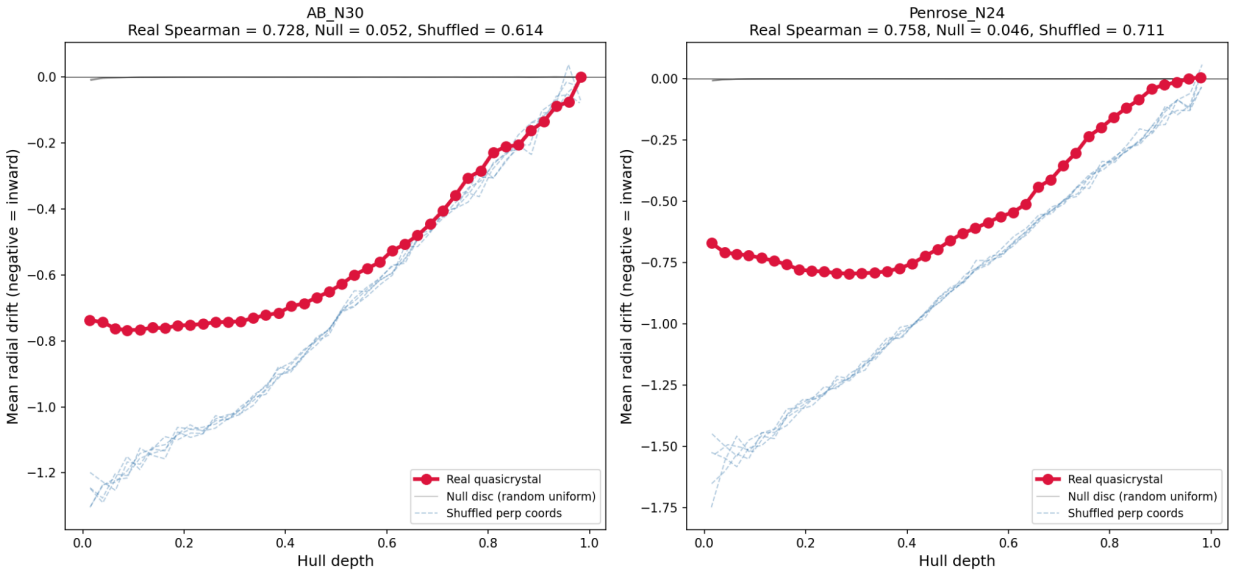


Figure 6: Depth-dependent directional balance in real tilings, null-disc controls, and shuffled-perpendicular controls. The null disc shows effectively zero correlation - the quasicrystal structure, not boundary geometry, is the source of the directional-balance effect.

The null disc produces a Spearman correlation of 0.05 — trivial boundary geometry contributes almost nothing. The real tiling produces 0.73–0.76. The quasicrystalline projection adds everything.

The shuffled-perpendicular control — in which perpendicular-space coordinates are randomly reassigned while the graph topology is preserved — retains 84–94% of the signal (0.615–0.713). This indicates that the graph topology itself, which is a downstream consequence of the projection geometry, carries most of the directional-balance information. The perpendicular-space coordinates and the graph structure are not independent sources; they are two expressions of the same underlying projection, and the directional-balance effect is encoded in both.

7.3 Summary of the mechanism

The complete mechanistic picture is now established:

Every edge in a projection tiling displaces the walker by unit magnitude in perpendicular space. The directions of these unit displacements are distributed symmetrically at deep hull positions and asymmetrically at shallow ones, due to truncation of the direction set by the acceptance-window boundary. The asymmetry is radial and inward, with no tangential component. Over a random walk, symmetric directions cancel (low address residue); asymmetric directions accumulate inward drift (high address residue). This is confirmed as genuine quasicrystal structure by null-disc controls that show no directional-balance effect despite having identical boundary geometry.

The mechanism is geometric, instantaneous, and positional. It requires no dynamical process, no relaxation timescale, and no area dependence. It is a property of how the acceptance-window boundary shapes the direction sets of vertices at each hull depth.

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8. Discussion

The mechanism reported in §§5–7 — depth-dependent directional balance arising from acceptance-window boundary truncation — is the primary contribution of this paper. This section examines what the mechanism implies for the channel architecture introduced in prior work, corrects an earlier characterisation, and notes connections to broader questions.

8.1 Address and weave are independent channels

The directional-balance mechanism operates entirely within perpendicular space and is defined by the projection geometry. It is natural to ask whether it interacts with the graph-topological (weave) channel, or whether the two channels carry independent information.

A unified perturbation experiment provides a direct test. Two perturbations — degree-preserving rewiring (which scrambles the graph while preserving perpendicular-space coordinates) and perpendicular-space shuffling (which randomises the hidden coordinates while preserving the graph) — are applied to both substrates, and identity-persistence AUC is measured for each channel separately:

	Native	Rewired	Shuffled
Address	0.979 / 0.553	0.979 / 0.569	0.497 / 0.486
Weave	0.999 / 0.908	0.991 / 0.831	0.999 / 0.909

(AB / Penrose; interior 75% subset, 10 replicates.)

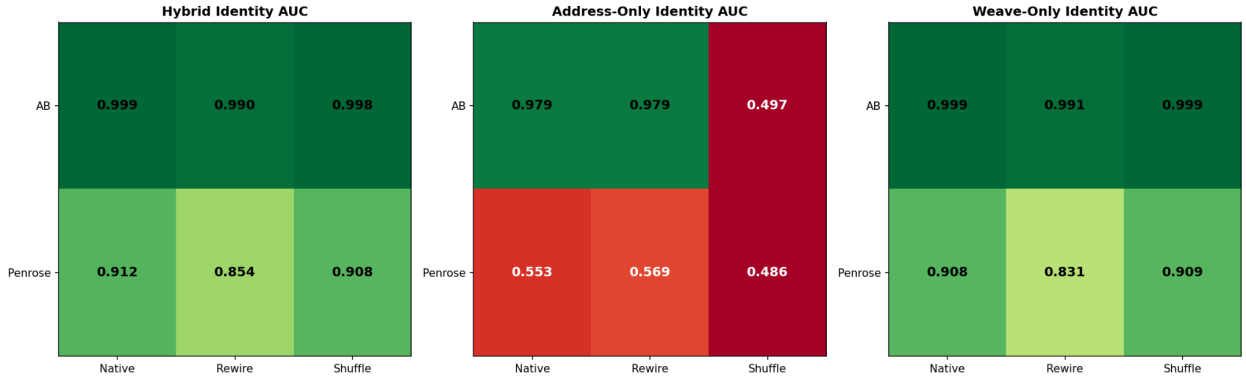


Figure 7: Channel-level perturbation responses. Shuffling perpendicular-space coordinates destroys the address channel (top row drops to chance) while leaving the weave channel intact. Rewiring the graph damages the weave channel while leaving the address channel intact. The two channels are independently targetable.

Shuffling kills address and preserves weave. Rewiring damages weave and preserves address. This is a channel-level double dissociation: two information channels, each carrying identity-relevant information, each independently destructible. The address channel encodes identity through projection coordinates that are invariant under graph modification; the weave channel encodes identity through graph-topological features that are invariant under coordinate reassignment.

8.2 Both substrates are weave-primary

The prior study (Niedzwiecki, 2026) reported that on the full AB patch, address AUC (0.874) exceeded weave AUC (0.740) after rewiring, and characterised AB as "address-led." This characterisation requires correction.

The comparison was between native address features and rewired weave features — an asymmetric test in which the weave channel had been subjected to the very perturbation designed to destroy it while the address channel had not. When both channels are measured on the unperturbed tiling — the fair comparison for assessing relative channel strength — weave exceeds address on both substrates, at every spatial scale:

Substrate	Subset	Address AUC	Weave AUC
AB	Full patch	0.84	0.98

Substrate	Subset	Address AUC	Weave AUC
AB	Interior 75%	0.98	1.00
AB	Boundary 25%	0.76	0.94
Penrose	Full patch	0.62	0.94
Penrose	Interior 75%	0.55	0.91
Penrose	Boundary 25%	0.64	0.95

Weave is the primary identity channel for both substrates. The "address-led" characterisation of AB does not survive measurement under matched conditions.

8.3 The exo/endo distinction is about backup, not leadership

If both substrates are weave-primary, what remains of the exo/endo distinction? The answer is visible when the primary channel is destroyed.

When rewiring eliminates the weave at the boundary — the regime where both channels are weakest and the directional-balance mechanism produces the strongest address drift — the two substrates diverge:

At boundary	Weave (rewired)	Address (intact)	Hybrid (rewired)
AB	0.528	0.762	0.776
Penrose	0.514	0.638	0.637

AB's address channel, though weaker than its weave, is strong enough to rescue the hybrid signal when weave fails (0.776 from a weave floor of 0.528). Penrose's address channel is too weak to compensate (hybrid 0.637 from a weave floor of 0.514 — no recovery).

This is the exo/endo distinction, reframed. It is not about which channel leads — both substrates are weave-led. It is about whether a working backup exists when the primary channel fails. AB has redundancy: two independent channels, either sufficient to carry identity at the boundary. Penrose has fragility: a single channel, with no parachute if the weave is destroyed.

The distinction matters precisely at the boundary, where the directional-balance mechanism produces the strongest perpendicular-space drift and where both channels are under the most geometric stress. The acceptance-window boundary is simultaneously where addresses are most unstable (§§6–7), where weave features weaken, and where the presence or absence of a backup channel becomes consequential.

8.4 The fossil/ecology split revisited

The prior study's strongest result — the fossil/ecology split between address and weave — is reinforced rather than undermined by the present findings. Address remains a fossil record: it preserves where structure was, survives destruction of the graph, and carries no information about the system's future organisation. Weave remains a living ecology: it determines where structure forms now and carries no memory of prior configurations.

What the present study adds is the mechanistic grounding for why the fossil record is unreliable at the boundary. The directional-balance mechanism shows that perpendicular-space addresses are geometrically unstable near the acceptance-window edge — not because the address is damaged or noisy, but because the direction sets available for perpendicular-space transport are truncated. The fossil is not eroded; it is written in a region where the recording medium itself is anisotropic.

8.5 Generality and connections

The directional-balance mechanism is a generic property of cut-and-project tilings. Any tiling constructed by projecting a higher-dimensional lattice will have uniform local perpendicular-space transport magnitude (all edges are projections of unit lattice bonds) and depth-dependent direction sets (the acceptance-window boundary truncates available directions). The specific geometry of the acceptance window — octagonal for AB, pentagonal for Penrose — determines the quantitative details, but the qualitative mechanism is universal to the construction.

Higher-dimensional source lattices would exhibit richer directional structure. The E8 lattice, with 240 nearest neighbours in 8 dimensions projecting to 6-dimensional perpendicular space, would provide a substantially larger direction set at each vertex, with correspondingly finer depth-dependent modulation of balance. Whether this produces qualitatively different behaviour — sharper transitions, additional structure in the directional-balance profile — is an open question.

A suggestive parallel exists with disorder-enhanced transport in physical quasicrystals. Levi et al. (2011) demonstrated three photonic transport regimes in quasicrystalline potentials: perfect quasicrystalline order traps light, moderate disorder frees it, and heavy disorder re-traps it via Anderson localisation. The directional-balance mechanism identifies the geometric basis for the first regime — the deep hull, where balanced directions produce cancellation and confine transport — and the transition to the boundary regime, where truncated directions permit drift. We note this as a structural correspondence, not an established connection.

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9. Limitations

This study is a classical graph-and-path analysis of finite tiling patches, not a quantum simulation. The holonomy proxy is a classical diagnostic — perpendicular-space displacement accumulated over random walks — and does not constitute a measurement of Berry phase, quantum holonomy, or any gauge-theoretic quantity. We use "holonomy-like" throughout to denote the structural parallel (physical closure without hidden-space closure) while maintaining the distinction from the formal objects of differential geometry and gauge theory.

The substrates are single finite patches generated by the de Bruijn multigrid method, not ensembles of independently generated tilings. Interior subsets control for boundary contamination but do not by themselves establish finite-size scaling. The depth-dependent directional balance is a geometric property of the projection construction and should hold at any patch size, but the quantitative details of the depth-balance profile — the precise Spearman values, the depth at which balance transitions from biased to symmetric — may shift with tiling scale. Systematic finite-size analysis is a natural next step.

The controlled linger experiment (§5) rules out temporal relaxation at the timescales tested (up to 160 steps). It does not rule out dynamical effects at substantially longer timescales or under walk protocols not examined here. We regard this as unlikely given the strength of the positional evidence, but note it as a logical gap.

The anomalous-diffusion exponent $\alpha \approx 0.553$ predicted in prior theoretical work (Niedzwiecki, 2025) was not recovered from any classical walk protocol tested in this study. Whether this prediction applies to a different transport regime (quantum, wave-mechanical, or non-Markovian) or requires revision remains open.

The connection between the directional-balance mechanism and disorder-enhanced transport in physical quasicrystals (§8.5) is noted as a structural correspondence. Establishing a quantitative link would require either wave-transport simulations on the tiling substrates or comparison with experimental photonic quasicrystal data, neither of which is attempted here.

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10. Conclusion

Aperiodic projection tilings exhibit a discrete holonomy-like effect: paths that nearly close in physical space can leave nonzero residual displacement in perpendicular space. This address residue is depth-gated — governed by the walker's position within the perpendicular-space acceptance window — and does not depend on loop area, dwell time at depth, or local step magnitude.

The mechanism is depth-dependent directional balance. Every edge in a projection tiling produces identical perpendicular-space displacement magnitude, but the directions available at each vertex vary with hull depth. Deep within the acceptance

window, direction sets are symmetric and random-walk contributions cancel; near the boundary, direction sets are truncated by the window edge and biased radially inward, so contributions accumulate as net address drift. This is confirmed as genuine quasicrystal structure — not trivial boundary geometry — by null-disc controls (Spearman 0.05 vs 0.73–0.76 for real tilings).

Both Ammann–Beenker and Penrose substrates are weave-primary for identity persistence, correcting the earlier characterisation of AB as address-led. The substrates differ not in which channel leads but in whether a working backup exists: when the weave is destroyed, AB's perpendicular-space address rescues the identity signal at the boundary; Penrose's cannot. The acceptance-window boundary — where directional balance is most biased and address drift is largest — is precisely the regime where this backup asymmetry becomes consequential.

We propose that directional-balance diagnostics and silent-corruption resistance be evaluated alongside erasure recoverability in the assessment of aperiodic substrates for error-correcting architectures.

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Data and code availability

All tiling datasets, analysis scripts, and reproducible code are available at https://github.com/kekekatie.